

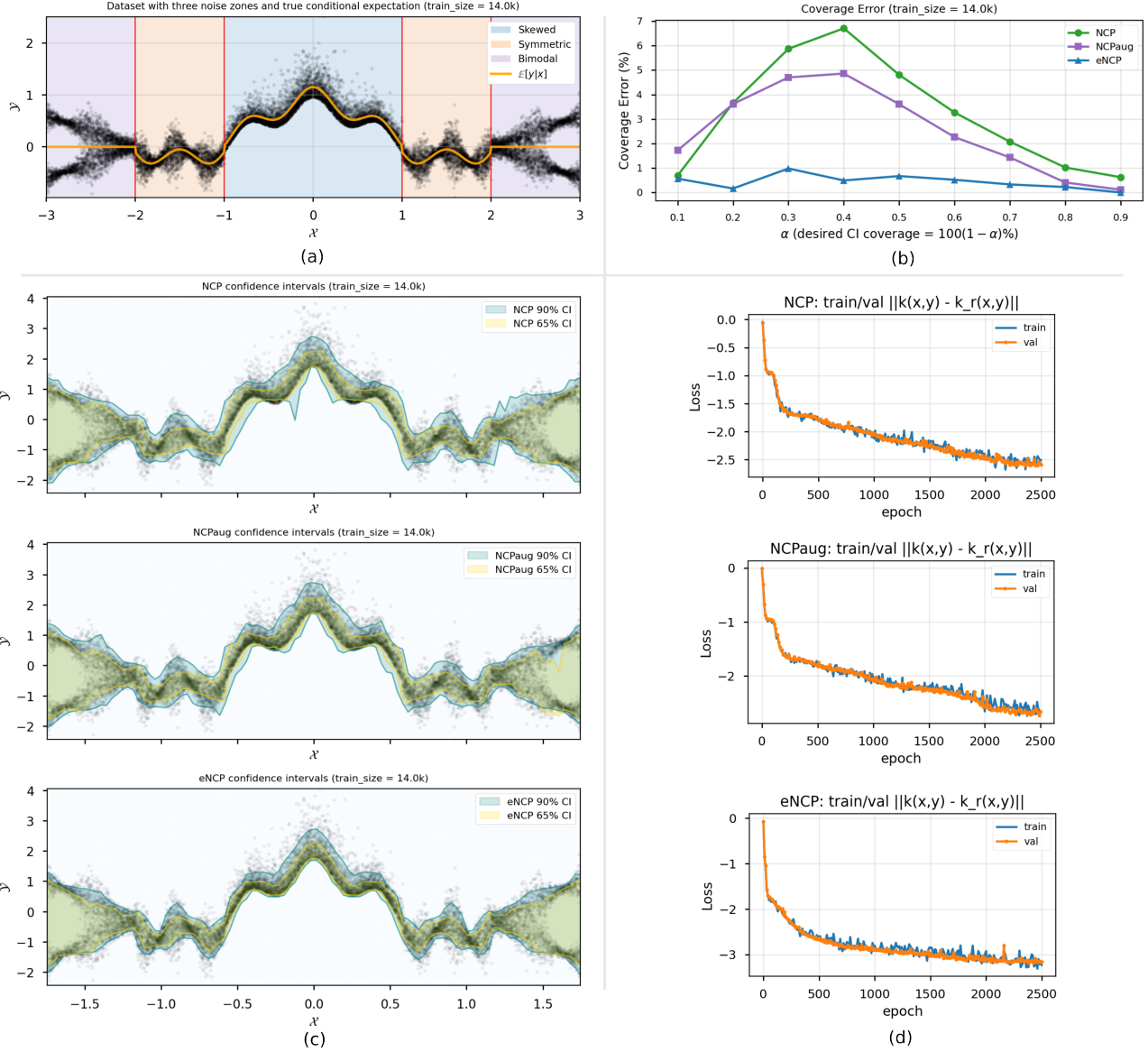


Figure A1. (a) Synthetic conditional distribution $P_{y|x}$ with three distinct zones whose conditional-distribution families share an uninformative conditional mean. Zone 1 ($|x| \leq 1$) is exponentially skewed, zone 2 ($1 < |x| \leq 2$) is symmetric, and zone 3 ($|x| > 2$) is bimodal. All zones include heteroscedastic noise. (b) Predicted confidence-interval coverage error for eNCP, NCP, and NCP with data augmentation (NCPaug) across desired coverage levels, defined as $100\% \cdot (1 - \alpha)$. (c) Predicted confidence intervals at desired coverages of 90% and 65% for NCP, NCPaug, and eNCP. (d) Training curves of train and validation losses for NCP, NCPaug, and eNCP.

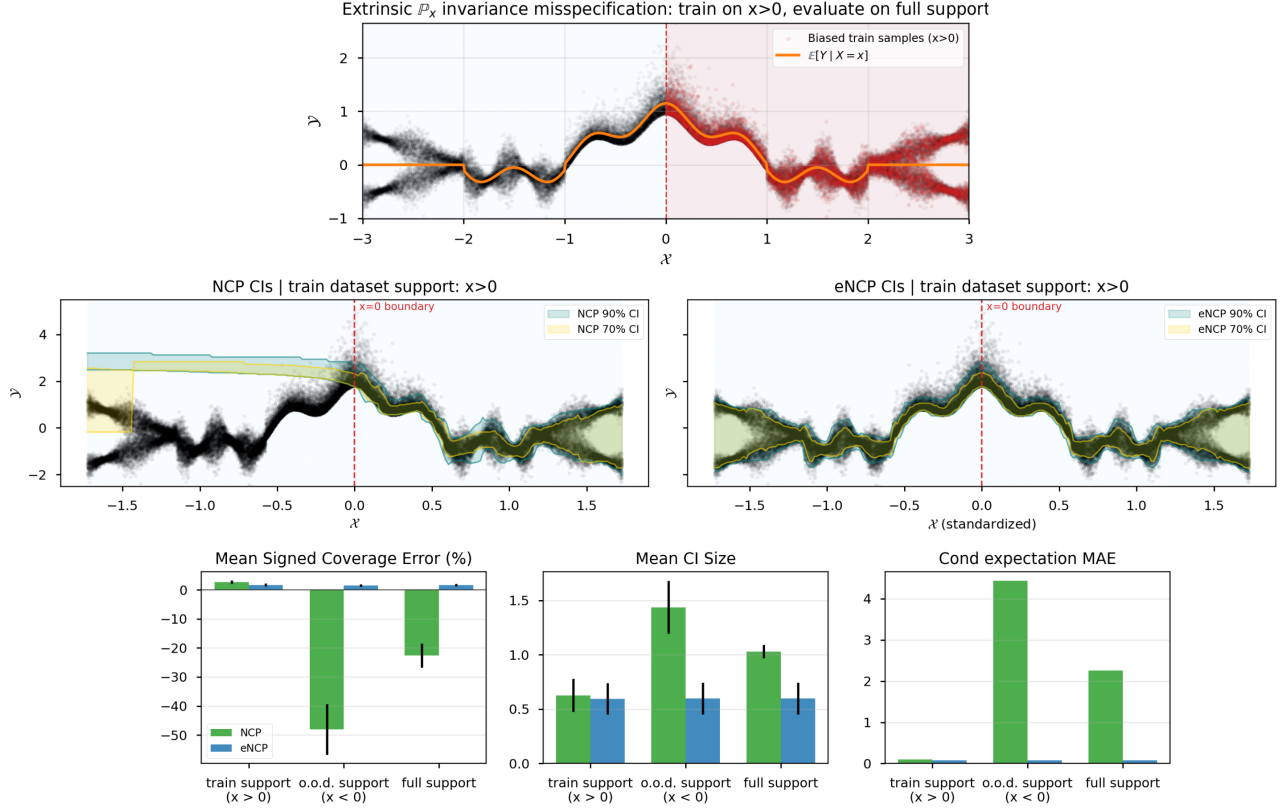


Figure A2. Top: Illustration of extrinsic symmetry misspecification, where the train/validation/test support is biased toward the coset subspace $x > 0$, violating the assumed $\mathbb{C}_2$ -invariance of P_X . Middle: Predicted confidence intervals at 90% and 70% coverage for NCP and eNCP under extrinsic symmetry misspecification. Bottom: Mean coverage error, confidence-interval volume, and conditional-expectation Maximum Absolute Error (MAE) for NCP and eNCP, evaluated on the biased support $x > 0$, the out-of-distribution (o.o.d.) support $x < 0$, and the full support $\mathcal{X} \equiv \mathbb{R}$.

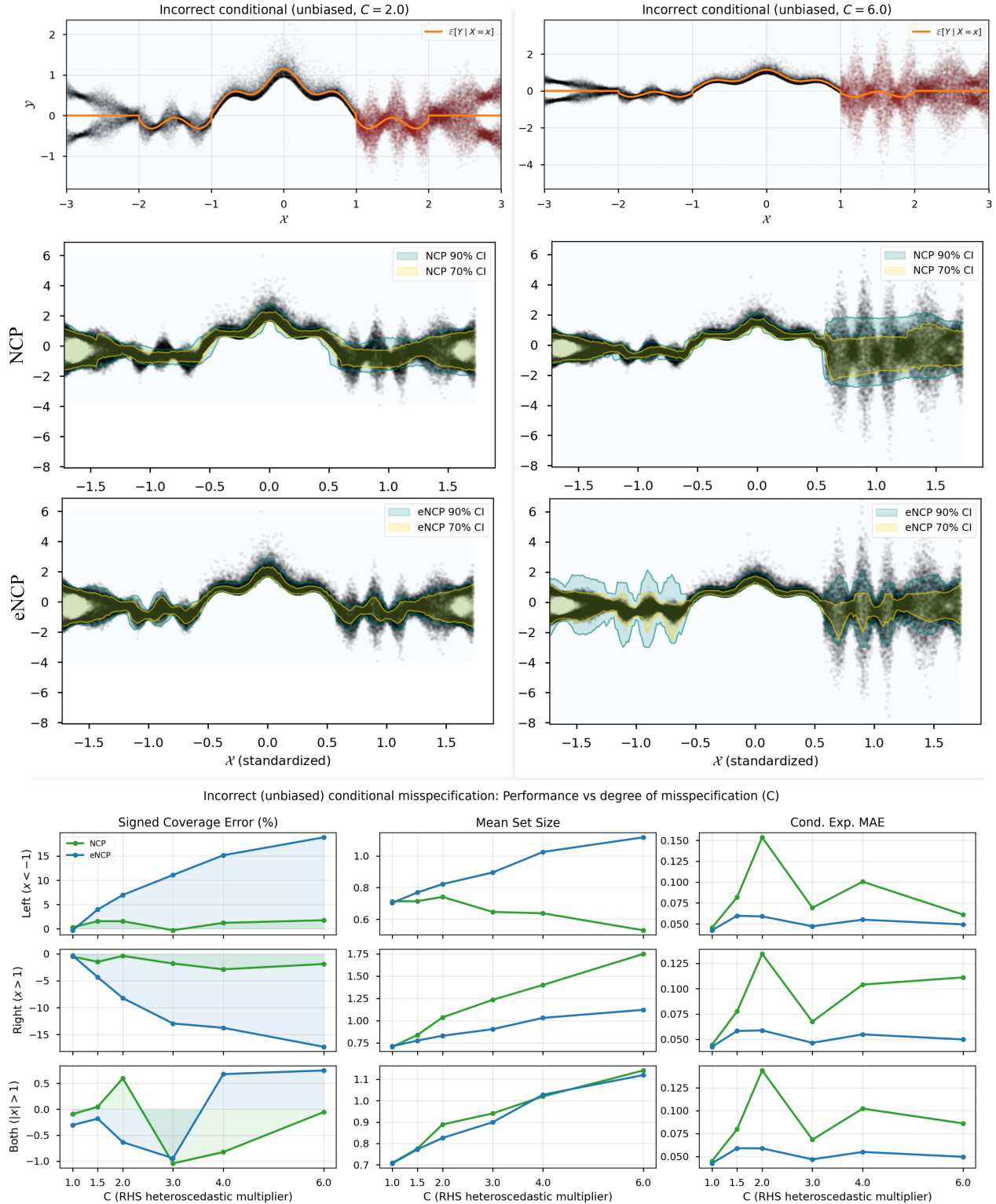


Figure A3. First row: Illustration of incorrect (unbiased) conditional misspecification. The $\mathbb{C}_2$ -invariance of $P_{Y|X}$ is violated on $|x| > 1$ by increasing the heteroscedastic noise scale with $C \in [1.0, 6.0]$ only on the right-coset subspace $x > 1$. This misspecification is termed unbiased because the conditional expectation $E[Y|X = \cdot]$ remains $\mathbb{C}_2$ -equivariant. Second and third rows: Predicted confidence intervals at 90% and 70% coverage for **NCP** and **eNCP** with $C = 2.0$ and $C = 6.0$ under this incorrect unbiased conditional misspecification. Bottom: Mean (signed) coverage error, confidence-interval volume, and conditional-expectation Maximum Absolute Error (MAE) for **NCP** and **eNCP**, evaluated on the misspecified supports $x > 1$ and $x < -1$, and on the full support $X \equiv \mathbb{R}$, across noise scales $C \in [1.0, 6.0]$.

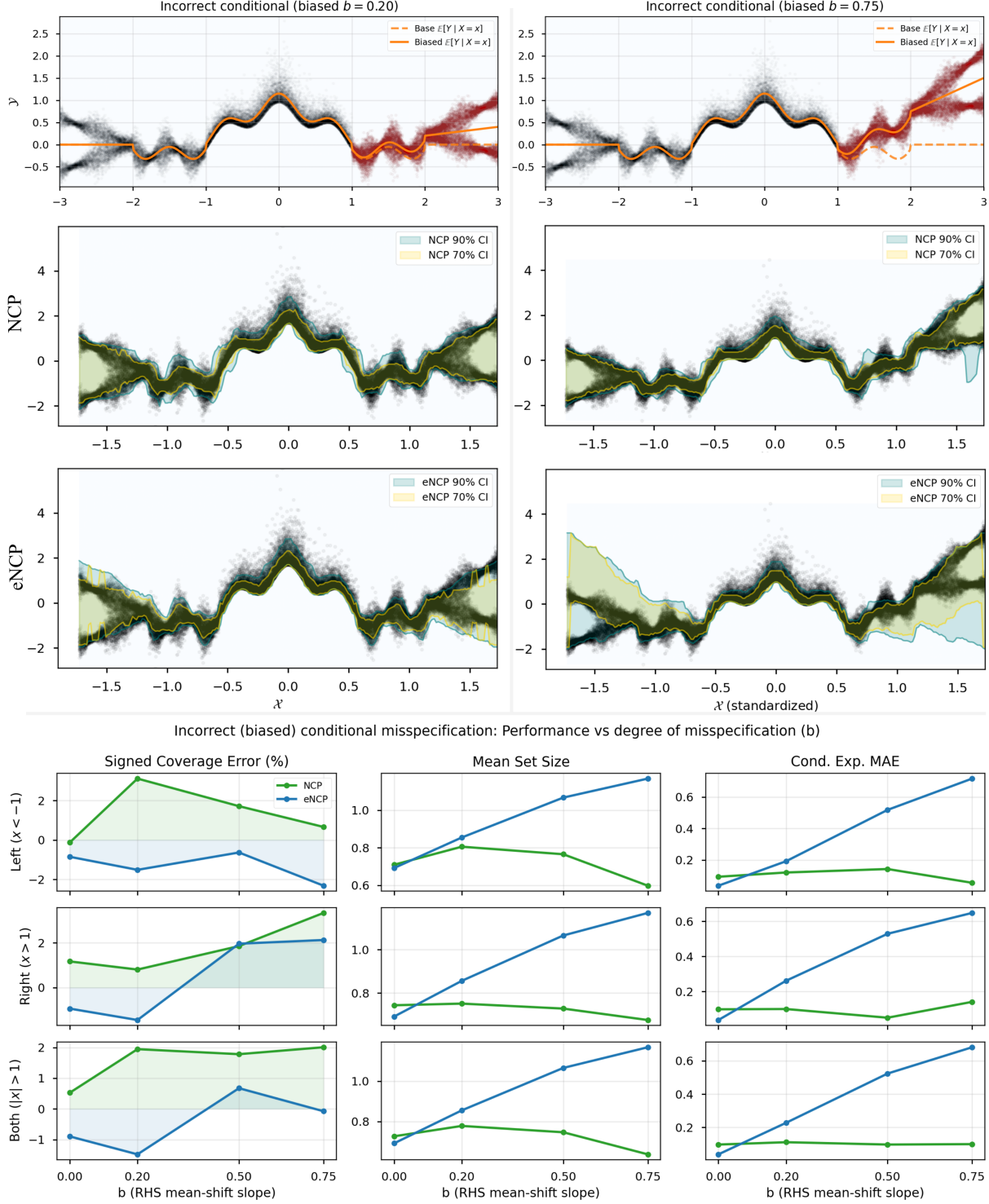


Figure A4. First row: Illustration of incorrect (biased) conditional misspecification. The $\mathcal{C}_2$ -invariance of $P_{y|x}$ is violated on $|x| > 1$ by introducing a linear bias with slope $b \in [0.0, 0.75]$ on the right-coset subspace $x > 1$. This misspecification is termed biased because the conditional expectation $\mathbb{E}[y | x = \cdot]$ is no longer $\mathcal{C}_2$ -equivariant. Second and third rows: Predicted confidence intervals at 90% and 70% coverage for NCP and eNCP with $b = 0.2$ and $b = 0.75$ under this incorrect biased conditional misspecification. Bottom: Mean (signed) coverage error, confidence-interval volume, and conditional-expectation Maximum Absolute Error (MAE) for NCP and eNCP, evaluated on the misspecified supports $x > 1$ and $x < -1$, and on the full support $\mathcal{X} \equiv \mathbb{R}$, across bias slopes $b \in [0.0, 0.75]$.